

C Source Code

```
static u64 micro_rotl64(u64 value,
u32 shift)
{
    shift &= 63U;
    if (shift == 0) {
        return value;
    }
    return (value << shift)
        | (value >> (64U - shift));
}
```

x86-64 (Optimal)

```
movl %esi, %ecx
movq %rdi, %rax
rolq %cl, %rax
```

BPF Bytecode

clang

```
r1 &= 63
if r1 == 0 goto +6
r2 = 64
r2 -= r1
r3 = r0
r3 <<= r1
r0 >>= r2
r0 |= r3
```

x86-64 (Under Optimized)

```
and $0x3f, %rdi
test %rdi, %rdi
je <done>
mov $0x40, %esi
sub %rdi, %rsi
mov %rax, %rdx
push %rcx
mov %rdi, %rcx
shl %cl, %rdx
pop %rcx
push %rcx
mov %rsi, %rcx
shr %cl, %rax
pop %rcx
or %rdx, %rax
```

JIT